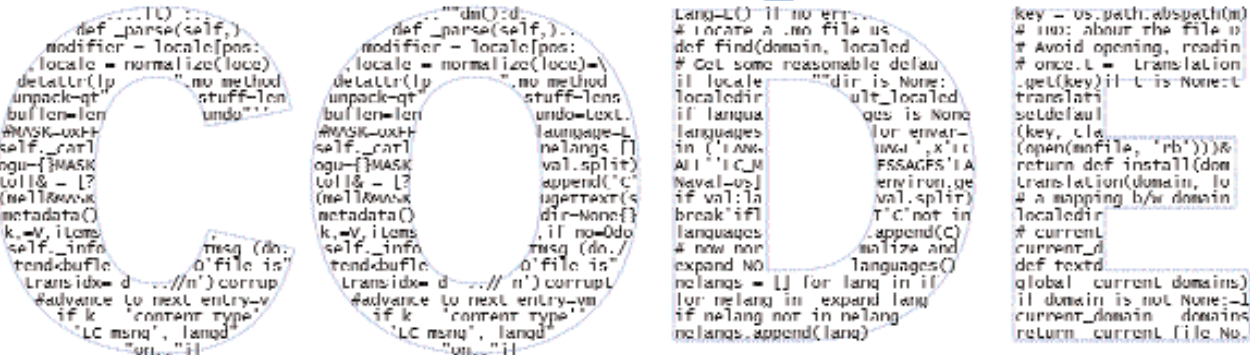


tracking the



Concurrent Versioning System is a unique concept that lets multiple programmers track the simultaneous changes during the working of the project.

Programming remains a dark art for those of us who have never dabbled in it. Even people who have done some programming—say, a short course in Visual Basic at a neighbourhood computer training institute—are well aware of the huge gap between amateur programs and the full-scale commercial software applications.

Isn't it a mystery to you how something as complex as an operating system, which requires scores of people working in tandem for long periods of time, is created? Programmers depend on certain specific techniques to create those wonderful applications which perform magic on your desktop. Besides the trusted text editor and the all-powerful compiler, professional programmers also depend on version control tools to help them with their jobs.

Déjà vu

CVS or Concurrent Versioning System generally refers to an ability to create and manage different versions of the same file. This concept is of prime use in environments where a lot of people work simultaneously on one big programming project, and they require the ability to store previous versions of the files. Developers can record the history of their source files and avoid major problems caused by bugs creeping into software during its long and complicated development cycle. Often, when a large group is working simultaneously on a project, bugs get introduced into the system which might not be detected until much

later when the modifications have already been made. Also, what happens frequently is that when a developer makes a change that fixes a minor problem related to functionality, he or she could end up creating a completely new set of problems that cause the application to cease functioning. Version control is particularly handy in these kinds of situations because it allows you to revert back to the version which works. Plus, having earlier versions helps in case there's a loss of data, and it provides you with a trackable record of the work that has been done in the past.

One of the more common (and basic) examples of version control is the versioning feature in Microsoft Word. It allows you to save multiple versions within a single Word document. The software also allows you to review the differences between two versions of the document using a feature that highlights only the changes that have been made.

Another form of version control is provided by the ARJ archival format. Users of this software can save multiple versions of any type of file—not just Word documents. Chapter archives in ARJ are very useful for backing up frequently modified file sets. You can create a sequence of snapshots of a directory that changes over time and ARJ saves only the files that have changed. When you extract the contents, it recreates the complete file set as it was at that

point in time. Since each chapter archive can contain up to 250 chapters, this means considerable space savings when compared to the usual practice of using multiple separate archives for backup.

Working with the codebase	
The programmer	The action that occurs at the CVS server
	
Views the file	A copy of the file is created <i>File is still available to other developers</i>
	
Modifies the file	The file is checked out <i>{File is locked and no other developer can overwrite it}</i>
	
Updates the file	The file is checked in <i>{The original and modified versions are compared and the server copy is updated}</i>
	
Commits changes	File is checked in <i>{The modified file is ready and waiting for the next developer to work on}</i>
	

ILLUSTRATION: Jaya Shetty